

课后习题..

第0章:

→ B取A的第2,4行,第1,4,6列

07. 设 $A = (a_{ij})_{4 \times 7}$, 其子矩阵 $B = \begin{bmatrix} a_{21} & a_{24} & a_{26} \\ a_{41} & a_{44} & a_{46} \end{bmatrix}$, 求C,D, 使 $B = CAD$.

解: 记单位矩阵的基列为 e_i , 则

$$\begin{bmatrix} e_1^T \\ e_4^T \end{bmatrix} A = \begin{bmatrix} e_1^T \\ e_4^T \end{bmatrix} \begin{bmatrix} A_1 \\ A_4 \end{bmatrix} = \begin{bmatrix} A_2 \\ A_4 \end{bmatrix} \quad (1)$$

$$\text{又有 } \begin{bmatrix} A_2 \\ A_4 \end{bmatrix} \cdot [e_1 \ e_4 \ e_6] = \begin{bmatrix} a_{21} & a_{24} & a_{26} \\ a_{41} & a_{44} & a_{46} \end{bmatrix} = B$$

$$\text{由(1)(2)得: } \begin{bmatrix} e_1^T \\ e_4^T \end{bmatrix} \cdot A \cdot [e_1 \ e_4 \ e_6] = B.$$

$$\text{可得: } C = \begin{bmatrix} e_1^T \\ e_4^T \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, D = [e_1 \ e_4 \ e_6] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

09. (1) 记 $e = e_1 + e_2 + \dots + e_n$, $Ae = (s_1, s_2, \dots, s_n)^T$, 求 $s_i (1 \leq i \leq n)$;

(2) 已知 n 阶方阵 A 的每行元素和为 a , 求 A^k 的每行元素和为 a^k , k 为正整数, 且 A 为可逆.

以上命题对 $k=-1$ 也成立. $Ae = a \cdot e$

$$\text{解: (1) } e = \sum_{i=1}^n e_i = [1, 1, \dots, 1]^T$$

设 $A = (a_{ij})$, 则 $s_i = \sum_{j=1}^n a_{ij}$ 为 A 的第 i 行元素之和.

(2) 由(1)可知, A^k 的每行元素之和为 $A^k e$ 的对应元素, 记为 t_i^k .

$$\text{又 } A^k \cdot e = A^{k-1} (A \cdot e) = A^{k-1} \cdot a \cdot e = a \cdot A^{k-1} \cdot e$$

$$\therefore t_i^k = t_i^{k-1} \cdot a = t_i^{k-2} \cdot a^2 = \dots = a^k$$

即 k 为正整数时, A^k 每行元素和为 a^k .

当 A 为可逆时, 有 $e = (A^{-1} A) \cdot e = A^{-1} \cdot (Ae) = A^{-1} (ae) = a \cdot A^{-1} e$

$$\therefore A^{-1} e = \frac{1}{a} e = a^{-1} e \quad \text{即 } A^{-1} \text{ 每行元素和为 } a^{-1}.$$

012. (1) 已知 A, B 满足 $A+B=AB$, 证 $A-I$ 可逆, 且求逆;

(2) 已知 $A^2=A$, 证 $A-2I$ 可逆, 且求逆.

证: (1) 由 $A+B=AB$ 得 $AB-A-B=0$

$$\text{则 } (A-I)(B-I)=I.$$

$\therefore (A-I)$ 与 $(B-I)$ 互为逆矩阵, $(A-I)^{-1} = B-I.$

(2) 由 $A^2=A$, $A \cdot A - A = 0$ $(A-2I)(A+I) = 2I.$

$$\therefore (A-2I) \cdot \left[\frac{1}{2} (A+I) \right] = I.$$

$\therefore A-2I$ 与 $-\frac{1}{2}(A+I)$ 互为逆矩阵, $(A-2I)^{-1} = -\frac{1}{2}(A+I)$

22.

15. 试证: 秩为 r 的矩阵可分解为一个秩为 1 的矩阵之和. (ppt 31)

证: 设该矩阵 A 的秩为 r , 则可化为 $I_{r \times r}$ 形式, 则

$$A_{m \times n} = P_{m \times m} I_{r \times r} Q_{r \times n} \quad \text{其中 } P, Q \text{ 可逆.}$$

$$\text{由 } I_{r \times r} = \sum_{k=1}^r I_{(k)} \quad \text{其中 } I_{(k)} = (a_{ij})_{r \times r}$$

$$a_{ij} = \begin{cases} 1 & i=j=k \\ 0 & \text{else.} \end{cases}$$

$$\text{则 } A_{m \times n} = \sum_{k=1}^r P I_{(k)} Q, \text{ 其中}$$

$$r(P \cdot I_{(k)} Q) \leq \min\{r(P), r(I_{(k)}), r(Q)\} \leq 1$$

$$\text{又 } r(P \cdot I_{(k)} Q) \geq r(P) + r(I_{(k)}) + r(Q) - m - n = 1$$

$$\therefore r(P \cdot I_{(k)} Q) = 1$$

$\therefore$ 原命题成立.

16. 设 A 为 n 阶方阵, B 是秩为 r 的 $n \times n$ 矩阵 (称为行满秩). 试证:

(1) 若 $AB=0$, 则 $A=0$

(2) 若 $AB=B$, 则 $A=I$.

$$(1) \text{ 由 } r(AB) \geq r(A) + r(B) - n$$

$$\text{得 } 0 \geq r(A) + r - n$$

$$\therefore r(A) = 0$$

$$\therefore A=0$$

$$(2) \text{ 由 } AB=B \text{ 得 } AB-B=0$$

$$\therefore (A-I)B=0$$

$$\text{由 (1) 得 } A-I=0$$

$$\therefore A=I.$$

8. 求下列矩阵的满秩分解.

$$(1) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \quad (2) \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 4 \end{bmatrix}$$

$$\text{解 (1)} \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

$$(2) \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

21. 试证: 幂等阵 A (即 $A^2=A$) 有 $r(A) + r(I_n - A) = n$

$$\text{由 } A^2=A \text{ 得 } A^2-A=0, A(A-I_n)=0$$

$$\therefore r[A(I_n-A)]=0$$

$$\therefore r[A(I_n-A)] \leq r(A) + r(I_n-A) - n.$$

$$\therefore r(A) + r(I_n-A) \leq n. \quad (1)$$

$$\text{又 } r(A) + r(I_n-A) \geq r[A + (I_n-A)] = n. \quad (2)$$

$$\text{由 (1), (2) 得 } r(A) + r(I_n-A) = n.$$

22. 试证: 若 n 阶方阵 A 满足 $r(A) + r(I_n - A) = n$, 则 $A^2=A$.

$$\text{证: 证 } A^2=A \text{ 即 } r(A^2-A)=0$$

又

$$\text{由 } r(A) + r(I_n - A) = n$$

$$\text{令矩阵 } P = \begin{bmatrix} A & I_n - A \end{bmatrix}, \text{ 则}$$

$$r(P) = r(A) + r(I_n - A) = n$$

$$\text{设 } Q = \begin{bmatrix} I_n & I_n \\ A & I_n - A \end{bmatrix} \cdot P \cdot \begin{bmatrix} I_n & A \\ I_n & A \end{bmatrix}$$

$$= \begin{bmatrix} A & I_n - A \\ A^2 & A - A^2 \end{bmatrix} \begin{bmatrix} I_n & A \\ I_n & A \end{bmatrix}$$

$$= \begin{bmatrix} I_n & 0 \\ 0 & A - A^2 \end{bmatrix}$$

$$\therefore r(Q) = r(I_n) + r(A - A^2) \geq n \quad (1)$$

$$\text{又 } r(Q) \leq r(P) = n$$

$$\therefore \text{由 (1), (2) 得 } r(Q) = n.$$

$$\therefore r(A - A^2) = 0, \therefore A - A^2 = 0$$

25. 设 n 阶方阵 $B_1, \dots, B_k$ 满足 $\sum_{i=1}^k B_i = 0$, 试证:

$$\sum_{i=1}^k r(B_i) \leq (k-1)n$$

$$\text{证: } 0 = r(B_1 + B_2 + \dots + B_{k-1} + B_k) \geq r(B_1 + \dots + B_{k-1}) + r(B_k)$$

$$\geq r(B_1 + \dots + B_{k-1}) + r(B_k)$$

$$\geq r(B_1) + \dots + r(B_k)$$

$$\geq r(B_1) + \dots + r(B_k) - (k-1)n$$

第1章
下列集合是实数域上的线性空间. 证明之.

(1) $V = \{ (n_1, n_2, \dots, n_k) \mid n_i \text{ 为整数, 数域为实数域 } R, \text{ 加法与数乘为通常运算} \}$

(2) $V = \{ (x_1, x_2) \mid x_1, x_2 \in R, \text{ 数域为 } R, \text{ 加法 } \oplus \text{ 与数乘 } \otimes \text{ 定义为:}$

$$(x_1, x_2) \oplus (y_1, y_2) = (x_1 + 2y_1, x_2 + 2y_2)$$

$$k \otimes (x_1, x_2) = (kx_1, 2kx_2)$$

(3) $V = R^+$, 数域为 R , 加法 $\oplus$ 与数乘 $\otimes$ 定义为:

$$a \oplus b = ab$$

$$k \otimes a = a^k$$

(4) $V = R^2 = \{ (x_1, x_2) \mid \forall x_i \in R \}$, 加法 $\oplus$ 与数乘 $\otimes$

$$\text{定义为: } (x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1, x_2 + y_2 + 2y_1)$$

$$k \otimes (x_1, x_2) = (kx_1, kx_2 + \frac{k(k-1)}{2} x_1^2) \quad (\forall k, x_1, y_1 \in R)$$

证: 要证明是线性空间. 即证 V 关于 $\oplus$ 与 $\otimes$ 满足线性空间的 8 条公理.

(1) $\because$ 当 k 为任意实数时 $\forall x \in V, kx \in V$. $\therefore$ 关于数乘封闭. 即证线性空间的 1 条公理.

$\therefore$ 不是线性空间

(2) V 不是线性空间. 因为加法不满足交换律.

$$(y_1, y_2) \oplus (x_1, x_2) = (y_1 + x_1, y_2 + x_2 + 2x_1) \neq (x_1, x_2) \oplus (y_1, y_2)$$

(3) V 是线性空间. 证明如下:

$$A_1: a \otimes b = ab = ba = b \otimes a \quad (\text{满足交换律})$$

$$A_2: (a \otimes b) \otimes c = ab \otimes c = abc = a \otimes (bc) = a \otimes bc = abc \quad (\text{满足结合律})$$

$$A_3: \exists 1 \in R^+, \forall a \in R^+, a \otimes 1 = a \cdot 1 = a. \quad (1 \otimes a = a)$$

$$A_4: \forall a \in R^+, \exists a^{-1} \in R^+, \text{ 使得 } a \otimes a^{-1} = a \cdot a^{-1} = 1$$

$$M_1: \forall a \in R^+, 1 \otimes a = a' = a$$

$$M_2: \forall k, l \in R, \forall a \in R^+, k \otimes (l \otimes a) = k \otimes a^l = a^{kl} = a^{lk} = (l \otimes a)^k = l \otimes a^k = l \otimes a^k$$

$$\begin{aligned} M_3: \forall k, l \in R, \forall a \in R^+ & (k \otimes l) \otimes a = (kl) \otimes a \\ & = a^{kl} = a^k \cdot a^l \\ & = a^k \otimes a^l \\ & = (k \otimes a^l) \otimes a \end{aligned}$$

$$M_4: \forall k \in R, \forall a, b \in R^+$$

$$k \otimes (a \otimes b) = k \otimes (ab) = (ab)^k = a^k \cdot b^k = (k \otimes a) \otimes (k \otimes b)$$

3. 证: 若 $\alpha_1, \dots, \alpha_r$ 线性无关, $\alpha_1, \dots, \alpha_r, \beta$ 线性相关, 则 β 可由 $\alpha_1, \dots, \alpha_r$ 线性表示, 且系数唯一.

证: $\alpha_1, \dots, \alpha_r, \beta$ 线性相关

$$\therefore \exists \text{ 不全为 } 0 \text{ 的数 } k_1, k_2, \dots, k_r, k \text{ 使得 } k_1 \alpha_1 + \dots + k_r \alpha_r + k \beta = 0$$

$$\text{若 } k = 0, \text{ 则 } k_1 \alpha_1 + \dots + k_r \alpha_r = 0$$

且 $k_1, \dots, k_r$ 不全为 0, 与 $\alpha_1, \dots, \alpha_r$ 线性无关矛盾

$$\therefore k \neq 0$$

$\therefore \beta$ 可由 $\alpha_1, \dots, \alpha_r$ 线性表示

$$\beta = (-\frac{k_1}{k}) \alpha_1 + \dots + (-\frac{k_r}{k}) \alpha_r$$

证系数唯一.

证: 设系数不唯一. 又两组系数 $a_1, \dots, a_r$ 和 $b_1, \dots, b_r$ 使得

$$\beta = a_1 \alpha_1 + \dots + a_r \alpha_r$$

$$\beta = b_1 \alpha_1 + \dots + b_r \alpha_r$$

两式相减得:

$$(a_1 - b_1) \alpha_1 + \dots + (a_r - b_r) \alpha_r = 0$$

$\therefore \alpha_1, \dots, \alpha_r$ 线性相关.

$$\therefore a_i = b_i, \dots, a_r = b_r$$

$\therefore \beta$ 可由 $\alpha_1, \dots, \alpha_r$ 唯一线性表示.

4. 求下列线性空间的维数及一组基.

(1) $F^{n \times n}$ 中全体对称阵所构成 F 上线性空间

(2) $C^{n \times n}$ 中全体上三角阵所构成 C 上线性空间

(3) $V(F) = \{ (x_1, x_2, \dots, x_{n+1}, x_n) \mid x_2 = x_4 = \dots = x_n, \forall x_i \in F \}$

取 $A = \text{diag}(1, w, w^2, \dots, w^{n-1})$, 其中 $w^3 = 1$ 但 $w \neq 1$, 且

$$V(R) = \{ f(A) \mid f(x) \in R[x] \}$$

解: $C^{n \times n}$ 全体上三角阵可经 $E_{ij} \quad (1 \leq i \leq j \leq n)$ 线性表示. 且 $\{E_{ij} \mid 1 \leq i \leq j \leq n\}$ 线性无关. 因此是线性空间的一组基. 维数为 $\frac{n(n+1)}{2}$

(2) $V(F)$ 可由 $f_1^T, f_2^T, \dots, f_{n+1}^T, (e_1^T + e_2^T + \dots + e_n^T)$ 线性表示. 且该向量线性无关. 故是 $V(F)$ 的一组基. 其维数为 $n+1$.

5. 设 $A \in C^{n \times n}$

(1) 若 $V = \{B \mid AB = BA, B \in C^{n \times n}\}$, 证 V 为 $C^{n \times n}$ 的子空间

(2) 若 $A = I$, 求 (1) 问中的 V

(3) 若 $A = \text{diag}(1, 2, \dots, n)$, 求 (1) 问中的 V 的基

(4) 当 $n=3$, $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, 求 (1) 问中的 V 的基

解: (1) 证子空间, 即证 V 关于 C 数域上线性运算封闭.

$\forall B, D \in V, \forall k \in F, kB + lD \in V$.

$\therefore V$ 关于 $C^{n \times n}$ 在数域 F 上线性运算封闭

$\therefore V$ 是 $C^{n \times n}$ 的子空间

(2) 当 $A = I$, 则 $V = \{B \mid B = B, B \in C^{n \times n}\}$, 即 $B \in C^{n \times n}$ 满足.

$\therefore V = C^{n \times n}$.

(3) 由 $AB = BA$, 得

$$\begin{bmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nn} \end{bmatrix} = \begin{bmatrix} b_{11} & 2b_{12} & \dots & nb_{1n} \\ b_{12} & 2b_{22} & \dots & nb_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & 2b_{n2} & \dots & nb_{nn} \end{bmatrix}$$

$\therefore$ 若 $i \neq j, b_{ij} = 0$ B 为对角矩阵.

$\therefore V$ 的一组基为 $\{E_{ij} \mid i=1, 2, 3, \dots, n\}$

(4) 由 $AB = BA$ 得

$$\begin{bmatrix} 3b_{11} & 3b_{12} & 3b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31}+b_{21} & b_{32}+b_{22} & b_{33}+b_{23} \end{bmatrix} = \begin{bmatrix} 3b_{11} & b_{21}+b_{13} & 2b_{13} \\ 3b_{12} & b_{22}+b_{23} & 2b_{23} \\ 3b_{31} & b_{32}+b_{23} & 2b_{33} \end{bmatrix}$$

$$\text{得 } B = \begin{bmatrix} b_{11} & 0 & 0 \\ 0 & b_{22} & 0 \\ 0 & b_{32} & b_{22}+b_{33} \end{bmatrix}$$

$\therefore B$ 可以线性表示为

$$b_{11}E_{11} + b_{22}(E_{22} + E_{33}) + b_{32}(E_{32} + E_{33})$$

$\therefore V$ 的一组基为 $\{E_{11}, E_{22}+E_{33}, E_{32}+E_{33}\}$.

7. 已知 $C^{2 \times 2}$ 的子空间:

$$V_1 = \left\{ \begin{bmatrix} x & y \\ y & x \end{bmatrix} \mid x, y \in C \right\}$$

$$V_2 = \left\{ \begin{bmatrix} x & y \\ x & y \end{bmatrix} \mid x, y \in C \right\}$$

分别求 $V_1 \cap V_2$ 及 $V_1 + V_2$ 的基.

解: ① $V_1 \cap V_2 = \left\{ \begin{bmatrix} x & x \\ x & x \end{bmatrix} \mid x \in C \right\}$.

$\therefore V_1 \cap V_2$ 的基为 $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

②. 由 ① $\dim(V_1 \cap V_2) = 1$.

$$\therefore \dim(V_1 + V_2) = \dim(V_1) + \dim(V_2) - \dim(V_1 \cap V_2)$$

易得 V_1 的基为 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ 及 $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

V_2 的基为 $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ 及 $\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$

$\therefore V_1 = \langle A, B \rangle, V_2 = \langle C, D \rangle$

$\therefore V_1 + V_2 = \langle A, B, C, D \rangle$

又 $\dim(V_1 + V_2) = 3$.

$\therefore V_1 + V_2$ 的基取 A, B, C, D 中线性无关的

3 个作为 $V_1 + V_2$ 的基. ~~即 A, B, C~~ .

$$\text{对应自然基下坐标} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

8. 设 $V(R)$ 为一切实连续函数所构成的线性空间作 $V(R)$ 的子空间:

$$V_1 = \{f(x) \mid f(-x) = f(x)\} \text{ (偶函数)}$$

$$V_2 = \{f(x) \mid f(-x) = -f(x)\} \text{ (奇函数)}$$

试证: $V = V_1 \oplus V_2$.

解: 证明直和. 即证 ① $V_1 + V_2$ 是直和 (即 $V_1 \cap V_2 = \{0\}$) ② $V = V_1 + V_2$

$$\text{① } V_1 \cap V_2 = \{f(x) \mid f(x) = f(x) = -f(x)\}$$

$$= \{f(x) \mid f(x) = 0\}$$

$$= \{0\}$$

$\therefore V_1 + V_2$ 是直和.

$$\text{② } f(x) = \underbrace{f(x) + f(-x)}_{\in V_1} + \underbrace{f(x) - f(-x)}_{\in V_2}$$

即 $\forall f(x) \in V$, 都可以写成偶函数与奇函数的和

$$\therefore V \subseteq V_1 + V_2. \text{ 又 } V_1 + V_2 \subseteq V.$$

$$\therefore V_1 + V_2 = V$$

证毕

10. 设 $V_1 = \{ (x_1, \dots, x_n) \mid x_1 + \dots + x_n = 0 \}$,
 $V_2 = \{ (x_1, \dots, x_n) \mid x_1 = x_2 = \dots = x_n \}$

试证: $C^n = V_1 \oplus V_2$.

证: ① $V_1 \cap V_2 = \{ (x_1, \dots, x_n) \mid x_1 + \dots + x_n = 0, x_1 = x_2 = \dots = x_n \}$
 $= \{ (x_1, \dots, x_n) \mid x_1 = x_2 = \dots = x_n = 0 \}$
 $= \{0\}$

② $V_1 + V_2$ 为直和

②. 首先 $V_1 + V_2 \subseteq C^n$
 $\forall \eta \in C^n, \eta = (x_1, \dots, x_n) \mid x_i \in C$
 $\exists \alpha = \frac{1}{n}(x_1 + \dots + x_n) \in V_2$

$\eta_1 = \eta - \alpha(1, \dots, 1) \in V_1$

$\eta_2 = \alpha(1, \dots, 1) \in V_2$

$\eta = \eta_1 + \eta_2$

$\therefore C^n \subseteq V_1 + V_2$

由①②得证.

11. 设 $A, B \in F^{n \times n}$, 且 $AB=0, B^2=B$. 又
 $V_1 = \{X \mid AX=0, X \in F^n\}, V_2 = \{X \mid BX=0, X \in F^n\}$

试证: (1) $F^n = V_1 + V_2$.

(2) $F^n = V_1 \oplus V_2 \Leftrightarrow r(A) + r(B) = n$.

证: (1) $\forall X \in F^n$, 有

$X = BX + (X - BX)$

由 $AB=0 \Rightarrow A(BX)=0 \therefore BX \in V_1$

由 $B^2=B \Rightarrow B(X-BX) = BX - B^2X = 0 \therefore X-BX \in V_2$

$\therefore F^n \subseteq V_1 + V_2$ 又 $V_1 + V_2 \subseteq F^n$

$\therefore F^n = V_1 + V_2$

(2) 对 V_1 有 $\dim(V_1) = n - r(A)$

V_2 有 $\dim(V_2) = n - r(B)$

① $\Rightarrow$ $\because F^n = V_1 \oplus V_2$

$\therefore \dim(F^n) = n = \dim(V_1) + \dim(V_2)$

$n = n - r(A) + n - r(B)$

$\therefore r(A) + r(B) = n$.

② $\Leftarrow \because r(A) + r(B) = n \therefore$
 $\therefore \dim(F^n) = n = 2n - (r(A) + r(B))$
 $= n - r(A) + n - r(B)$
 $= \dim(V_1) + \dim(V_2)$

可得 $F^n = V_1 \oplus V_2$

由①②可得 $F^n = V_1 \oplus V_2 \Leftrightarrow r(A) + r(B) = n$.

13. 在 $F^{2 \times 2}$ 中定义线性变换 $f(X) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} X$,
 $\forall X \in F^{2 \times 2}$, 分别求 f 在基 $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ 下
基 $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ 下的矩阵.

解: $f(E_{11}) = \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} f(E_{12}) = \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix}$

$f(E_{21}) = \begin{bmatrix} 0 & b \\ 0 & d \end{bmatrix} f(E_{22}) = \begin{bmatrix} 0 & b \\ 0 & d \end{bmatrix}$

设 f 在基 $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ 下矩阵为 A .
在基 $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ 下矩阵为 B

则根据由线性变换可得

$A = \begin{bmatrix} a & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ c & 0 & 0 & 0 \\ 0 & c & 0 & 0 \end{bmatrix} B = \begin{bmatrix} a & b & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & b & d \\ 0 & 0 & c & d \end{bmatrix}$

$f(E_{11}, E_{12}, E_{21}, E_{22}) = (E_{11}, E_{12}, E_{21}, E_{22}) A$

$f(E_{11}, E_{12}, E_{21}, E_{22}) = (E_{11}, E_{12}, E_{21}, E_{22}) B$

14. 设线性变换 f 在基 $\varepsilon_1, \varepsilon_2, \varepsilon_3$ 下矩阵为 $A = (a_{ij})_{3 \times 3}$.

(1) 求 f 在基 $\varepsilon_1, \varepsilon_2, \varepsilon_3$ 下的矩阵.

(2) 求 f 在基 $\varepsilon_1 + k\varepsilon_2, \varepsilon_2, \varepsilon_3$ 下的矩阵.

解: 由题意得

① $(\varepsilon_1, \varepsilon_2, \varepsilon_3) = (\varepsilon_1, \varepsilon_2, \varepsilon_3) P$
 $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} P^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = P$

$\therefore f$ 在 $\varepsilon_1, \varepsilon_2, \varepsilon_3$ 下的矩阵为:

$B = P^{-1}AP = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

②. $[\varepsilon_1 + k\varepsilon_2, \varepsilon_2, \varepsilon_3] = [\varepsilon_1, \varepsilon_2, \varepsilon_3] Q$

$Q = \begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} Q^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -k & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\therefore f$ 在 $\varepsilon_1 + k\varepsilon_2, \varepsilon_2, \varepsilon_3$ 下的矩阵为 $Q^{-1}AQ$

$$Q^T A Q = \begin{bmatrix} a_{11} + k a_{12} & a_{12} & a_{13} \\ a_{21} + k a_{22} - k a_{12} & a_{22} & a_{23} - k a_{13} \\ a_{31} + k a_{32} & a_{32} & a_{33} \end{bmatrix}$$

6. 证明下列映射是线性映射，并求基变换求线性映射矩阵。

(1) 证明 $f(A) = \text{tr} A \cdot I_n, \forall A \in R^{n \times n}$. $f: R^{n \times n} \rightarrow R$.

(2) $R[x]_3 = \{a_0 + a_1 x + a_2 x^2 \mid a_i \in R, h(x) = x^3 + x\}$

且 $f[p(x)] = \int_0^1 p(t) h(x+t) dt$ ($\forall p(x) \in R[x]_3$).

解: $\forall A, B \in R^{n \times n}$ $f(kA + lB) = \text{tr}(kA + lB)$
 $= k \cdot \text{tr} A + l \cdot \text{tr} B$
 $= k \cdot f(A) + l \cdot f(B)$

是线性映射

以 $\{E_{11}, E_{12}, \dots, E_{nn}, E_{21}, \dots, E_{nn}\}, \{1\}$ 为基
 $f(E_{ii}) = 1, f(E_{ij}) = 0, (i \neq j)$

设 f 的矩阵为 A , 则

$$[\underbrace{1, \dots, 1}_{ng}, \underbrace{0, \dots, 0}_{n^2-n}]_{1 \times n^2} = [1] A$$

$$A = [\underbrace{1, \dots, 1}_{ng}, \underbrace{0, \dots, 0}_{n^2-n}]_{1 \times n^2}$$

②. $\forall p(x), q(x) \in R_3[x]$ 有:

$$f(kp(x) + lq(x)) = \int_0^1 (kp(t) + lq(t)) h(x+t) dt$$

$$= k \int_0^1 p(t) h(x+t) dt + l \int_0^1 q(t) h(x+t) dt$$

$$= k f(p(x)) + l f(q(x))$$

是线性映射

以 $1, x, x^2$ 为基. $R_3[x]$ 的基. $p(t) = 1$

$$f(1) = \int_0^1 x^2 + tx dt = \left(\frac{1}{3} x^3 + \frac{1}{2} x^2 \right) \Big|_0^1 = \frac{1}{3} x + \frac{1}{2} x^2$$

$$f(x) = \int_0^1 x^3 + tx^2 dt = \frac{1}{2} x^2 + \frac{1}{3} x^3$$

$$f(x^2) = \int_0^1 x^4 + tx^3 dt = \frac{1}{5} x^5 + \frac{1}{4} x^4$$

$$f(x) = \int_0^1 t(x^2 + tx) dt = \int_0^1 t^2 x^2 + t^3 x dt = \frac{1}{5} x^2 + \frac{1}{4} x^3$$

$$f(x^2) = \int_0^1 t^2 x^2 + t^3 x dt = \frac{1}{4} x^2 + \frac{1}{5} x^3$$

$$[\frac{1}{3}x + \frac{1}{2}x^2, \frac{1}{2}x^2 + \frac{1}{3}x^3, \frac{1}{5}x^2 + \frac{1}{4}x^3] = [1, x, x^2] A$$

$$A = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{3} & \frac{1}{4} \end{bmatrix}$$

16. 分别求第15题中 f 的值域及核的一组基

解: f 的值域为 R . 基为 $\{1\}$.

f 的核为 $\{A \mid \text{tr} A = 0, A \in R^{n \times n}\}$

也可表示为 $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{n,n-1} & -(a_{11} + a_{22} + \dots + a_{n-1,n-1}) \end{bmatrix}$

取一组基为 $\{E_{ij} \mid 1 \leq i, j \leq n, E_{ii} - E_{nn} \mid k=1, 2, \dots, n-1\}$

(2) f 的值域为 $\{ax + bx^2 \mid a, b \in R\}$

基为 $\{x, x^2\}$

f 的核为 $\{p(x) \mid \int_0^1 p(t) h(x+t) dt = 0, \forall x \in R[x]_3\}$

$$\text{令 } p(t) = a_0 + a_1 t + a_2 t^2$$

$$a_0 x^2 + a_0 x + a_1 x^3 + a_1 x^2 + a_2 x^4 + a_2 x^3$$

$$\int_0^1 (a_0 x^2 + a_0 x + a_1 x^3 + a_1 x^2 + a_2 x^4 + a_2 x^3) dt$$

$$= a_0 x^2 + \frac{1}{2} a_0 x + \frac{1}{4} a_1 x^4 + \frac{1}{3} a_1 x^3 + \frac{1}{5} a_2 x^5 + \frac{1}{4} a_2 x^4$$

$$= a_0 x^2 + \frac{1}{2} a_0 x + \frac{1}{4} a_1 x^4 + \frac{1}{3} a_1 x^3 + \frac{1}{5} a_2 x^5 + \frac{1}{4} a_2 x^4$$

$$= a_0 x^2 + \frac{1}{2} a_0 x + \frac{1}{4} a_1 x^4 + \frac{1}{3} a_1 x^3 + \frac{1}{5} a_2 x^5 + \frac{1}{4} a_2 x^4$$

$$= \frac{1}{2} a_0 x + (a_0 + \frac{1}{2} a_1) x^2 + (a_1 + \frac{1}{3} a_2) x^3 + a_2 x^4$$

$$\int_0^1 (a_0 + a_1 t + a_2 t^2) (x^2 + tx) dt$$

$$= \int_0^1 (a_0 x^2 + a_0 tx + a_1 x^3 + a_1 tx^2 + a_2 x^4 + a_2 tx^3) dt$$

$$= [a_0 x^2 t + \frac{1}{2} a_0 x t^2 + \frac{1}{4} a_1 x^4 t + \frac{1}{3} a_1 x t^3 + \frac{1}{5} a_2 x^5 t + \frac{1}{4} a_2 x^4 t^2] \Big|_0^1$$

$$= a_0 x^2 + \frac{1}{2} a_0 x + \frac{1}{4} a_1 x^4 + \frac{1}{3} a_1 x^3 + \frac{1}{5} a_2 x^5 + \frac{1}{4} a_2 x^4$$

$$= a_0 x^2 + \frac{1}{2} (a_0 x + a_1 x^2) + \frac{1}{3} (a_1 x + a_2 x^2) + \frac{1}{4} a_2 x^3 = 0$$

$$(\frac{1}{2} a_0 + \frac{1}{3} a_1 + \frac{1}{4} a_2) x + (a_0 + \frac{1}{2} a_1 + \frac{1}{3} a_2) x^2 = 0$$

$$\therefore K(f) = \{p(x) \mid f(p(x)) = 0\}$$

$$6a_0 + 4a_1 + 3a_2 = -(12a_0 + 6a_1 + 4a_2)x$$

$$x = -\frac{6a_0 + 4a_1 + 3a_2}{12a_0 + 6a_1 + 4a_2}$$

17. 设 $f \in \text{Hom}(V, V)$

(1) 试证: f 是单射 $\Leftrightarrow K(f) = \{0\}$

(2) 若 $\dim V = n$, 试证: f 是单射 $\Leftrightarrow f$ 是满射 $\Leftrightarrow f$ 可逆

证: (1) $\Rightarrow$ 由 f 是单射, $\forall s_1, s_2 \in K(f)$ 有 $f(s_1) = f(s_2) = 0$

$$\text{则 } s_1 = s_2$$

即 $K(f)$ 只有一个元素

$$\text{又 } \{0\} \subseteq K(f) \text{ 故 } K(f) = \{0\}$$

$$\Leftarrow: \forall f(s_1) = f(s_2) \text{ 有 } s_1, s_2 \in V$$

$$\text{由线性映射 } f(s_1 - s_2) = f(s_1) - f(s_2) = 0$$

$$\therefore s_1 - s_2 \in K(f), \text{ 而 } s_1 - s_2 = 0, s_1 = s_2$$

$\therefore f$ 为单射

(2) $\Rightarrow$ 当 f 为单射, 由 $K(f) = \{0\}, \dim K(f) = 0$

$$\therefore \dim R(f) = \dim V - \dim K(f) = n$$

$$\therefore R(f) = V$$

$\therefore f$ 为满射

$$\Leftarrow: \text{当 } f \text{ 为满射 } \dim R(f) = \dim V = n$$

$$\therefore \dim K(f) = \dim V - \dim R(f) = 0$$

$$\therefore K(f) = \{0\}$$

由 (1) 可得 f 为单射

(2) $\Rightarrow$ 当 f 为满射可得 f 是单射

$\therefore f$ 为双射, $\therefore f$ 可逆

$\Leftarrow$: 当 f 可逆, 则 f 为双射, 故 f 为满射

$\therefore f$ 为满射 $\Leftrightarrow f$ 可逆

18. 设 $f \in \text{Hom}(V, V), \dim V = n$ 且 $f^2 = f$, 试证: f

的矩阵必相似于 $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}_{n \times n}$, 其中 $r = \dim R(f)$

证: 设 $x \in V$, 则 $x = f(x) + x - f(x)$

$$f(f(x)) = f^2(x) = f(x) \in R(f)$$

$$f(x - f(x)) = f(x) - f^2(x) = f(x) - f(x) = 0$$

$$\therefore x - f(x) \in K(f)$$

$$\text{又 } \dim R(f) + \dim K(f) = \dim V$$

$$R(f) + K(f) = V$$

$$\therefore V = R(f) + K(f)$$

取 $R(f)$ 中基: $f(\alpha_1), f(\alpha_2), \dots, f(\alpha_r)$

取 $K(f)$ 中基: $\beta_1, \beta_2, \dots, \beta_{n-r}$

则 $f(\alpha_1), f(\alpha_2), \dots, f(\alpha_r), \beta_1, \dots, \beta_{n-r}$ 为 V 的一组基

$$f^2(\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_{n-r})$$

$$= (f^2(\alpha_1), \dots, f^2(\alpha_r), 0, \dots, 0)$$

$$= (f(\alpha_1), \dots, f(\alpha_r), 0, \dots, 0)$$

$$= (f(\alpha_1), \dots, f(\alpha_r), \beta_1, \dots, \beta_{n-r}) \cdot A$$

$$\text{其中 } A = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

即 f 的矩阵为 A

当 f 取其它基与上述基的过渡矩阵为 P

则 f 的矩阵为 $B = P^{-1}AP$

所以在任意基下, f 的矩阵必相似于 $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

20. 已知线性变换 f 与 g 满足 $f^2 = f \cdot g^2 = g$. 试证:

(1) f 与 g 有相同的值域 $\Leftrightarrow fg = g, gf = f$.

(2) f 与 g 有相同核 $\Leftrightarrow fg = f, gf = g$.

证: ~~设 $x \in V$, 则 $x = f(x) + x$~~

(1) $\Rightarrow$: 设 $g(u) \in R(g)$

~~由 $f^2 = f$~~ $\exists \alpha$ 使 $g(u) = f(\alpha)$

有 $fg(u) = f \cdot f(\alpha) = f^2(\alpha) = f(\alpha) = g(u)$

$\therefore fg = g$. 同理 $gf = f$

$\Leftarrow$: $\forall \eta \in R(f) \exists \alpha$ 使得 $f(\alpha) = \eta$

$\therefore \eta = f(\alpha) = g(f(\alpha)) \in R(g)$.

同理 $\forall \eta' \in R(g)$ 有 $\eta' \in R(f)$.

$\therefore f$ 与 g 有相同的值域

~~证 (1) 得证.~~

(2) $\Rightarrow$: $\forall \eta \in V, \eta = f(u) + \eta - f(u)$

$\therefore f(\eta - f(u)) = f(\eta) - f^2(u) = 0$

$\therefore \eta - f(u) \in K(f)$

又 $K(f) = K(g) \therefore \eta - f(u) \in K(g)$

$\therefore g(\eta - f(u)) = 0$

$\therefore g(\eta) = g(f(u) + \eta - f(u))$

$= g(u) + g(\eta - f(u))$

$= g(u)$

$\therefore gf = g$ 同理 $fg = f$

$\Leftarrow$: $\forall \eta \in K(f)$ 则 $f(\eta) = 0$

$g(u) = g(u) = 0$

又 $gf = g \therefore g(\eta) = g(f(\eta)) = 0$

$\therefore \eta \in K(g)$

同理 $\forall \eta' \in K(g)$ 有 $\eta' \in K(f)$

$\therefore f$ 与 g 有相同的核

证得证.

第一章

2. 设 α 为线性空间 $V(F)$ 中非零向量, 若于中数 k_1 与 k_2 不相容, 试证: $k_1\alpha \neq k_2\alpha$.

证: (只要证明 $k_1\alpha \neq k_2\alpha$)
(只要证明 $k_1\alpha - k_2\alpha \neq 0$)

$$k_1\alpha - k_2\alpha = (k_1 - k_2)\alpha$$

$$\text{若 } (k_1 - k_2)\alpha = 0$$

$$\therefore \alpha = 0$$

$\therefore k_1 - k_2 = 0$ 与条件 $k_1 \neq k_2$ 不符.

$$\therefore k_1\alpha - k_2\alpha \neq 0$$

$$\text{即 } k_1\alpha \neq k_2\alpha$$

4. 在下列线性空间的维数及一组基.

(1) $F^{n \times n}$ 中全体对称阵所构成 F 上线性空间.

(2) $C^{n \times n}$ 中全体上三角阵所构成 C 上线性空间.

(3) $V(F) = \{ (x_1, x_2, \dots, x_{n-1}, x_n) \mid x_1 = x_2 = \dots = x_n, \forall x_i \in F \}$

(4) $A = \text{diag}(1, \omega, \omega^2)$, 其中 $\omega^3 = 1$, 但 $\omega \neq 1$. 且

$$V(F) = \{ f(A) \mid \exists f(x) \in R[x] \}$$

解: 基的定义.

$V(F)$ 大前提 $\alpha_1, \dots, \alpha_n \in V$.

1. 线性无关

2. $\forall \alpha \in V$, 可以由 $\alpha_1, \dots, \alpha_n$ 线性表示.

$$(1) \left\{ \begin{pmatrix} x & z \\ z & y \end{pmatrix} \mid x, y, z \in F \right\}$$

$$\begin{pmatrix} x & z \\ z & y \end{pmatrix} = x \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + z \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + y \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\text{基为 } \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$(2) \begin{pmatrix} x & z \\ 0 & y \end{pmatrix} = x \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + z \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + y \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$(3) F^{2n} \text{ 中 } (1, 0, \dots, 0) e_1$$

$$(0, 0, 1, \dots, 0) e_2$$

$$(0, 0, \dots, 1, 0) e_{2n-1}$$

$$(0, 1, 0, \dots, 0) e_2 + e_4 + \dots + e_{2n}$$

$\left\{ \begin{matrix} n+1 \text{ 个向量} \end{matrix} \right\}$

(4). $A \in F^{2 \times 2}$. 求 $V(F)$ 的一组基.

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n + \dots$$

$$f(A) = a_0I + a_1A + a_2A^2 + \dots + a_nA^n + \dots$$

$$\begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ & \omega \end{pmatrix} \begin{pmatrix} 1 & \\ & \omega^2 \end{pmatrix} \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \dots$$

$\therefore$ 基为 I, A, A^2 .

$$xI + yA + zA^2 = 0 \Rightarrow \begin{cases} x+y+z=0 \\ x+\omega y+\omega^2 z=0 \\ x+\omega^2 y+\omega z=0 \end{cases} \Rightarrow \begin{cases} x=0 \\ y=0 \\ z=0 \end{cases}$$

$$\text{系数行列式 } \begin{vmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{vmatrix} \neq 0$$

(5) 设 $A \in C^{n \times n}$

(1) 若 $V = \{ B \mid AB = BA, B \in C^{n \times n} \}$ 证 V 为 $C^{n \times n}$ 的子空间

(2) 若 $A = I$, 求 (1) 问中 V .

(3) 若 $A = \text{diag}(1, 2, \dots, n)$, 求 (1) 问中 V 的一组基.

(4) 当 $n=3, A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, 求 (1) 问中 V 的一组基.

解: (1) 验证关于线性运算封闭

$$\forall B_1, B_2 \in V$$

$$A(B_1 + B_2) = AB_1 + AB_2 = B_1A + B_2A = (B_1 + B_2)A$$

$$\therefore B_1 + B_2 \in V$$

$$\therefore \forall B \in V, k \in C$$

$$A(kB) = kAB = kBA = (kB)A$$

$$\therefore kB \in V$$

$\therefore \dots$

$$(2) V = C^{n \times n}$$

$$(3) \text{ 令 } B = (b_{ij})_{n \times n}, AB = BA$$

$$\Leftrightarrow B \text{ 为对角阵}$$

$$\text{基为 } E_1, E_2, \dots, E_n$$

$$(4) \text{ 令 } B = (b_{ij})_{n \times n}, B \in V \Leftrightarrow AB = BA$$

$$AB = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$BA = \dots$$

$$\begin{cases} 2b_{12} = b_{13} \\ b_{13} = 0 \\ b_{21} = 2b_{31} \\ b_{22} = b_{33} \\ b_{23} = b_{32} \end{cases}$$

$$A = I + \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$B = \begin{pmatrix} b_{11} & 0 & 0 \\ 0 & b_{22} & 0 \\ 0 & b_{32} & b_{33} \end{pmatrix}$$

6. 已知 $\alpha_1 = (1, 2, 1, 0)$ $\alpha_2 = (1, 1, 1, 1)$

$\beta_1 = (2, -1, 0, 1)$ $\beta_2 = (1, -1, 3, 7)$

$V_1 = \text{span}\{\alpha_1, \alpha_2\}$ $V_2 = \text{span}\{\beta_1, \beta_2\}$

分别求 $V_1 + V_2$, $V_1 \cap V_2$ 的一组基.

解: $V_1 + V_2 = \langle \alpha_1, \alpha_2, \beta_1, \beta_2 \rangle$

$$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 1 & -1 & -1 \\ 0 & 1 & 1 & 7 \\ 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 3 & -5 & -3 \\ 0 & 1 & 1 & 7 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 7 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \therefore \text{基 } \alpha_1, \alpha_2, \beta_1, \beta_2$$

$$\dim(V_1 + V_2) = 3$$

$$\therefore \dim(V_1 \cap V_2) = 1$$

$$V_1 + V_2: \eta \in V_1 \cap V_2 \Rightarrow \exists k_1, k_2 \text{ s.t. } \eta = k_1 \alpha_1 + k_2 \alpha_2$$

$$\exists l_1, l_2 \text{ s.t. } \eta = l_1 \beta_1 + l_2 \beta_2$$

$$k_1 \alpha_1 + k_2 \alpha_2 = l_1 \beta_1 + l_2 \beta_2$$

$$\Rightarrow k_1 \alpha_1 + k_2 \alpha_2 - l_1 \beta_1 - l_2 \beta_2 = 0$$

$$\begin{cases} k_1 - k_2 - 2l_1 - l_2 = 0 \\ 2k_1 + k_2 + l_1 + l_2 = 0 \\ k_1 + k_2 - 3l_2 = 0 \\ k_2 - l_1 - 7l_2 = 0 \end{cases}$$

系数矩阵初等行变换找到基础解系

$$\rightarrow \text{简化阶梯形矩阵} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \text{基础解系} \begin{pmatrix} k_1 \\ k_2 \\ l_1 \\ l_2 \end{pmatrix} = l_2 \begin{pmatrix} -1 \\ 4 \\ -3 \\ 1 \end{pmatrix}$$

$$\therefore \eta = -\alpha_1 + 4\alpha_2 = -3\beta_1 + \beta_2$$

$$\therefore V_1 \cap V_2 = \langle k(-\alpha_1 + 4\alpha_2) \rangle = \langle -\alpha_1 + 4\alpha_2 \rangle$$

$$\therefore V_1 \cap V_2 \text{ 基为 } -\alpha_1 + 4\alpha_2 \text{ 或 } -3\beta_1 + \beta_2$$

8. 设 $V_1 = \{A \mid A^T = A\}$ $V_2 = \{A \mid A^T = -A\}$

试证: $C^{n \times n} = V_1 \oplus V_2$

证: $\forall A \in C^{n \times n}$

$$A = A^T - A$$

$$\therefore A = 0$$

$$\therefore V_1 \cap V_2 = \{0\}$$

$$\therefore V_1 + V_2 \text{ 为直和.}$$

① $\forall A \in C^{n \times n}$

$$A = A_1 + A_2$$

$$= \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

$$A_1 \in V_1, A_2 \in V_2, \therefore C^{n \times n} \subseteq V_1 + V_2$$

11. 设 $A, B \in F^{n \times n}$, 且 $AB = 0, B^2 = B$ 且

$$V_1 = \{x \mid Ax = 0, x \in F^n\}, V_2 = \{x \mid Bx = 0, x \in F^n\}$$

试证: (1) $F^n = V_1 + V_2$

$$(2) F^n = V_1 \oplus V_2 \Leftrightarrow r(A) + r(B) = n$$

证: (1) $\because V_1, V_2$ 都是 F^n 的子空间

$$\therefore F^n = V_1 + V_2$$

$$\therefore \eta \in F^n$$

$$\eta = B\eta + (\eta - B\eta)$$

$$\therefore AB\eta = 0, B(\eta - B\eta) = B\eta - B^2\eta = 0$$

$$\therefore B\eta \in V_1, \eta - B\eta \in V_2$$

$$\therefore F^n \subseteq V_1 + V_2$$

$$\therefore F^n = V_1 + V_2$$

(2) 由齐次线性方程组解空间的维数与自由未知量的个数. 即:

$$\dim V_1 = n - r(A), \dim V_2 = n - r(B)$$

$$\therefore \dim V_1 + \dim V_2 = 2n - (r(A) + r(B))$$

$$\Rightarrow F^n = V_1 \oplus V_2 \text{ 即 } V_1 + V_2 \text{ 是直和.}$$

$$\Leftrightarrow \dim F^n = \dim V_1 + \dim V_2$$

$$\Leftrightarrow n = 2n - (r(A) + r(B))$$

$$\Leftrightarrow r(A) + r(B) = n$$

13. 在 $F^{2 \times 2}$ 中定义线性变换 $f(x) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} x$ 且 $x \in F^{2 \times 1}$, 分别求 f 在基 $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ 与基 $\{E_{11}, E_{12}, E_{21}, E_{22}\}$ 下的矩阵.

$$\text{解: } f(E_{11}) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix}$$

$$f(E_{12}) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix}$$

$$f(E_{21}) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & 0 \\ d & 0 \end{bmatrix}$$

$$f(E_{22}) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & b \\ 0 & d \end{bmatrix}$$

$$\text{则 } f \text{ 在 } \{E_{11}, E_{12}, E_{21}, E_{22}\} \text{ 下的矩阵为 } \begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & 0 \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix}$$

$$f \text{ 在 } \{E_{11}, E_{12}, E_{21}, E_{22}\} \text{ 下的矩阵为 } \begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & 0 \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix}$$

其次在 V_1, V_2 中分别取一组基.

$$V_1: e_1 \cdots e_r$$

$$V_2: e_{r+1} \cdots e_n$$

$$\text{则 } V = e_1 \cdots e_r, e_{r+1} \cdots e_n.$$

设 f 在这组基 $e_1 \cdots e_n$ 下的矩阵为

$$(f(e_1) \cdots f(e_r), f(e_{r+1}) \cdots f(e_n)) = (e_1, \cdots, e_r, e_{r+1}, \cdots, e_n) \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

第2章

⑤ 已知

$$W = \{\alpha_1, \alpha_2, \dots, \alpha_5\}^T \left| \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix} (\alpha_1, \dots, \alpha_5)^T = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right.$$

求 $W^\perp$ 的一组标准正交基.

解: $A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 & 4 \end{bmatrix}$

$$W = K(A) \quad W^\perp = K(A)^\perp = R(A^H) = R(A^T).$$

$$\therefore W^\perp = \mathcal{L}\left(\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix}\right) = \mathcal{L}(\alpha_1, \alpha_2). \quad \alpha_1 \text{ 与 } \alpha_2 \text{ 线性独立.}$$

$$\therefore W^\perp \text{ 的一组基为 } \alpha_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}.$$

标准正交基: $\beta_1 = \alpha_1$

$$\beta_2 = \alpha_2 - \frac{\langle \alpha_2, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1$$

$$= \alpha_2 - \frac{14}{55} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} - \frac{4}{5} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}$$

直接算
更简单

$$A \rightarrow \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix} = B.$$

$$W = K(A) = K(B).$$

$$W^\perp = K(B)^\perp = R(B^H) = R(B^T).$$

$$W^\perp = \mathcal{L}(\alpha_1, \alpha_2) = \mathcal{L}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}\right)$$

$$\therefore W^\perp \text{ 的一组基为 } \alpha_1 = (1, 1, 1, 1, 1)^T, \alpha_2 = (0, 1, 2, 3, 4)^T$$

标准正交基: $\beta_1 = \alpha_1$

$$\beta_2 = \alpha_2 - \frac{\langle \alpha_2, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1$$

$$= \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} - \frac{5}{5} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\text{单位化: } \gamma_1 = \frac{\beta_1}{\|\beta_1\|} = \frac{1}{\sqrt{5}} (1, 1, 1, 1, 1)^T$$

$$\gamma_2 = \frac{\beta_2}{\|\beta_2\|} = \frac{1}{\sqrt{14}} (1, -1, 2, 3, 4)^T$$

① 设 f 是内积空间 V 上的变换, 若

$$\langle f(\alpha), f(\beta) \rangle = \langle \alpha, \beta \rangle \quad (\forall \alpha, \beta \in V)$$

试证: f 是线性变换. 因此, f 是正交变换.

需证: $\forall \alpha, \beta \in V, k \in \mathbb{F}$.

$$f(k\alpha + l\beta) = kf(\alpha) + lf(\beta).$$

$$f(k\alpha + l\beta) - kf(\alpha) - lf(\beta) = 0.$$

$$\text{取 } \langle f(k\alpha + l\beta) - kf(\alpha) - lf(\beta), f(k\alpha + l\beta) - kf(\alpha) - lf(\beta) \rangle = 0$$

利用内积的可加性与齐性.

$$\text{上式右边} = \langle f(k\alpha + l\beta), f(k\alpha + l\beta) \rangle - \langle f(k\alpha + l\beta), kf(\alpha) \rangle - \langle f(k\alpha + l\beta), lf(\beta) \rangle$$

保持内积不变.

$$= \langle k\alpha + l\beta, k\alpha + l\beta \rangle - k \langle k\alpha + l\beta, \alpha \rangle - l \langle k\alpha + l\beta, \beta \rangle = 0$$

$\therefore f$ 是线性变换. ...

⑦ 设 V 为欧氏空间, k 为实数, $f(\alpha) = \alpha - k \langle \alpha, w \rangle w$, $\|w\| = 1$. 求 f 是正交变换的充要条件. (解法)

本题有两种解法. 一种方法根据正交变换的定义

即: f 是线性变换

① 保持内积不变.

$$\forall \alpha, \beta \in V, \langle f(\alpha), f(\beta) \rangle \stackrel{?}{=} \langle \alpha, \beta \rangle$$

$$\langle \alpha - k \langle \alpha, w \rangle w, \beta - k \langle \beta, w \rangle w \rangle =$$

$$\langle \alpha, \beta \rangle - k \langle \beta, w \rangle \langle \alpha, w \rangle - k \langle \alpha, w \rangle \langle \beta, w \rangle$$

$$+ k^2 \langle \alpha, w \rangle \langle \beta, w \rangle$$

$$= \langle \alpha, \beta \rangle + k^2 \langle \alpha, w \rangle \langle \beta, w \rangle - 2k \langle \alpha, w \rangle \langle \beta, w \rangle$$

$$k^2 - 2k = 0 \Rightarrow k = 0 \text{ 或 } 2.$$

另一种解法利用正交变换的几何条件. (PPT 156 (3))

将 w 扩充成 V 的一组标准正交基.

$$\alpha_1 = w, \alpha_2, \alpha_3, \dots, \alpha_n.$$

$$f(w) = f(\alpha_1) = \alpha_1 - k \langle \alpha_1, w \rangle w = (1-k)w$$

$$f(w_j) = w_j \quad (j=2, \dots, n)$$

$$\therefore f \text{ 在标准正交基 } w_1, \dots, w_n \text{ 下的矩阵 } A = \begin{pmatrix} 1-k & & \\ & 1 & \\ & & \ddots \end{pmatrix}$$

$$A^T A = I \Rightarrow (1-k)^2 = 1 \Rightarrow k = 0 \text{ 或 } 2.$$

8. 设 f 是内积空间 V 的共轭变换, w 是 f 的不变子空间, 试证: $w^\perp$ 也是 f 的不变子空间

证: $\forall \alpha \in w^\perp, f(\alpha) \in w^\perp$

证: $\forall \alpha \in w^\perp, \beta \in w$

$$0 = \langle \alpha, \beta \rangle = \langle f(\alpha), f(\beta) \rangle$$

要证 $f(\alpha) \in w^\perp$

$\therefore \beta \in w, w$ 不变

$\therefore f(\beta) \in w$

$\therefore \langle f(\alpha), f(\beta) \rangle = 0$

~~另证: 利用正交性~~

双正交空间的定义: $\eta \in w, \eta \in w^\perp \Leftrightarrow \forall \beta \in w, \langle \eta, \beta \rangle = 0$

但上面只说明了 $f(\alpha)$ 与 w 中 $f(\beta)$ 这种正交的向量

并未说明 $f(\alpha)$ 与 w 中任意向量正交。

证 $f(\beta)$ 实际上是构成 w 的基 $f(w)$

要说明 $f(w) = w$, 即要说明 $f|_w$ 是满射。

在有限维的空间上, 单射的线性变换是满射

的必要条件是它是单射

即证 $\ker(f|_w) = \{0\}$

即 $\forall \eta \in w, f(\eta) = 0 \Rightarrow \eta = 0$

$$\therefore \langle f(\eta), f(\eta) \rangle = \langle \eta, \eta \rangle$$

$$f(\eta) = 0$$

$$\therefore \langle \eta, \eta \rangle = 0$$

$$\therefore \eta = 0$$

$\therefore \ker(f|_w) = \{0\}$

$\therefore f|_w$ 是单射 $\Rightarrow$ 满射

$\therefore f(w) = w$ 再倒回去...

$\forall \alpha \in w^\perp, \beta \in w$

$f(\alpha) \in w^\perp$

$\therefore w^\perp$ 是 f 的不变子空间

10. 设 A, B 都是正交阵, 且 $\det A \det B = 1$, 试证: $\det(A+B) = 0$

证: $A^T A = I, |A| = \pm 1$

$$A+B = A(I + A^T B) = A(I + A^T B)$$

$$= A(B^T + A^T)B$$

$$= A(A+B)^T B$$

$$\therefore |A+B| = |A(A+B)^T B| = |A| \cdot |(A+B)^T| \cdot |B|$$

$$\therefore |A| |B| = 1$$

$$\therefore |(A+B)^T| = |A+B| = -|A+B|$$

$$\therefore |A+B| = 0$$

12. 设 $\|w\| = 1$, 试证镜像变换

$$H(x) = x - 2\langle x, w \rangle w, \forall x \in \mathbb{C}^n$$

在基 $e_1, \dots, e_n$ 下矩阵为 $I - 2ww^H$, 因此不论 w 怎样, 单位向量, 总有 $\det(I - 2ww^H) = -1$

证: 考虑 $f: \mathbb{C}^n \rightarrow \mathbb{C}^n, A$

$$x \mapsto Ax$$

f 在基 $e_1, \dots, e_n$ 下矩阵是 A

$$H(x) = x - 2w^H x \cdot w$$

$$= x - 2(w \cdot \underbrace{w^H x}_{n \times n \text{ 矩阵}})$$

$$= (I - 2ww^H)x$$

$\therefore H$ 在基 $e_1, \dots, e_n$ 下矩阵为 $I - 2ww^H$

同一个线性变换在不同基下的矩阵相似

且相似矩阵的行列式相等

因此, 要找一组新基 ($\mathbb{C}^n$ 的基) 使 H 的矩阵

的行列式为 -1 . (p616). 解法: 将 w 扩充成 V 的基...

在标准正交基 $w, w_2, \dots, w_n$ 下矩阵为 $\begin{pmatrix} -1 & & \\ & 1 & \\ & & \ddots & \\ & & & 1 \end{pmatrix}$ $\therefore$ 行列式为 -1

$$\therefore |I - 2ww^H| = -1$$

第2章

1. 试证内积空间的“平行四边形定理”:

$$\|\alpha + \beta\|^2 + \|\alpha - \beta\|^2 = 2(\|\alpha\|^2 + \|\beta\|^2)$$

$$\text{证: } \|\alpha + \beta\|^2 + \|\alpha - \beta\|^2$$

$$= \langle \alpha + \beta, \alpha + \beta \rangle + \langle \alpha - \beta, \alpha - \beta \rangle$$

$$= \langle \alpha, \alpha \rangle + \langle \alpha, \beta \rangle + \langle \beta, \alpha \rangle + \langle \beta, \beta \rangle +$$

$$\langle \alpha, \alpha \rangle - \langle \beta, \alpha \rangle - \langle \alpha, \beta \rangle + \langle \beta, \beta \rangle$$

$$= 2\langle \alpha, \alpha \rangle + 2\langle \beta, \beta \rangle$$

$$= 2\|\alpha\|^2 + 2\|\beta\|^2$$

2. 试证欧氏空间“勾股定理”:

$$\alpha \perp \beta \Leftrightarrow \|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$$

并讨论该命题在酉空间是否成立.

$$\text{证: } \|\alpha + \beta\|^2 = \langle \alpha + \beta, \alpha + \beta \rangle = \langle \alpha, \alpha \rangle + \langle \beta, \beta \rangle + 2\langle \alpha, \beta \rangle$$

$$= \|\alpha\|^2 + \|\beta\|^2 + 2\langle \alpha, \beta \rangle$$

$$\therefore \langle \alpha, \beta \rangle = 0$$

$$\therefore \alpha \perp \beta$$

$$\Rightarrow \because \alpha \perp \beta$$

$$\therefore \langle \alpha, \beta \rangle = 0$$

$$\therefore \langle \alpha, \alpha \rangle + \langle \beta, \beta \rangle + 2\langle \alpha, \beta \rangle = \langle \alpha, \alpha \rangle + \langle \beta, \beta \rangle$$

$$\therefore \|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$$

3. 设 $f_1(x), f_2(x), \dots, f_n(x)$ 是 $[a, b]$ 上实连续函数, 试证:

$$|\int_a^b f_i(x) f_j(x) dx| \leq \max \int_a^b f_k^2(x) dx \quad (i, j = 1, 2, \dots, n)$$

证: 在酉空间

$$\|\alpha + \beta\|^2 = \langle \alpha, \alpha \rangle + 2\operatorname{Re} \langle \alpha, \beta \rangle + \langle \beta, \beta \rangle$$

$$\text{由 } \|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2 \text{ 可得}$$

$$\operatorname{Re} \langle \alpha, \beta \rangle = 0$$

故不一定要 $\langle \alpha, \beta \rangle = 0$ 即不一定有 $\alpha \perp \beta$.

$$\Rightarrow \because \alpha \perp \beta \therefore \langle \alpha, \beta \rangle = 0 \quad \langle \beta, \alpha \rangle = \overline{\langle \alpha, \beta \rangle} = 0$$

$$\therefore \|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2 \text{ 成立}$$

$$\therefore \text{在酉空间中 } \alpha \perp \beta \Rightarrow \|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$$

3. 令 $V = \{f_i(x)\}$, $\langle f_i(x), f_j(x) \rangle = \int_a^b f_i(x) f_j(x) dx$

易知 V 是酉空间, 由 C-B 不等式可得:

$$|\langle f_i(x), f_j(x) \rangle| \leq \sqrt{\langle f_i(x), f_i(x) \rangle} \sqrt{\langle f_j(x), f_j(x) \rangle}$$

$$|\int_a^b f_i(x) f_j(x) dx| \leq \sqrt{\int_a^b f_i^2(x) dx} \sqrt{\int_a^b f_j^2(x) dx} \leq (\max_k \int_a^b f_k^2(x) dx)$$

$$\therefore |\int_a^b f_i(x) f_j(x) dx| \leq \max_k \int_a^b f_k^2(x) dx$$

4. 设 $A = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 1 & -1 \end{bmatrix}$, 把 A 的列作为欧氏空间 R^3 的一组基, 按 Schmidt 正化方法求 R^3 的一组标准正交基, 由此求出正交阵 Q 及上三角阵 R , 使 $A = QR$.

$$\text{解: } \alpha_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \quad \alpha_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \alpha_3 = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$$\text{化 } \beta_1 = \alpha_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

$$\beta_2 = \alpha_2 - \frac{\langle \alpha_2, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{0}{2} \beta_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\beta_3 = \alpha_3 - \frac{\langle \alpha_3, \beta_1 \rangle}{\langle \beta_1, \beta_1 \rangle} \beta_1 - \frac{\langle \alpha_3, \beta_2 \rangle}{\langle \beta_2, \beta_2 \rangle} \beta_2$$

$$= \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} - \frac{3}{2} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} - \frac{-3}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

$$\text{单位化: } \gamma_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}^T$$

$$\gamma_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{3} \end{pmatrix}^T$$

$$\gamma_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{3} \\ -\frac{\sqrt{3}}{6} \\ -\frac{\sqrt{3}}{6} \end{pmatrix}^T$$

$$\therefore \text{正交阵为 } Q = \begin{pmatrix} 0 & \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{6} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{3} & -\frac{\sqrt{6}}{6} \end{pmatrix}$$

$$R = \begin{pmatrix} \|\beta_1\| & \|\beta_1\| b_{12} & \|\beta_1\| b_{13} \\ 0 & \|\beta_2\| & \|\beta_2\| b_{23} \\ 0 & 0 & \|\beta_3\| \end{pmatrix} = \begin{pmatrix} \sqrt{2} & 0 & -\frac{3\sqrt{2}}{2} \\ 0 & \sqrt{3} & \frac{\sqrt{3}}{2} \\ 0 & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$$

8. 设 f 是内积空间 V 的共轭变换, W 是 f 的不变子空间, 试证: $W^\perp$ 也是 f 的不变子空间

证: 任取 $\alpha \in W^\perp$ 有 $f(\alpha) = ?$

$$\therefore \|f(\alpha)\| = \|\alpha\| = 0$$

$$\therefore \alpha = 0, \quad K(f) = \{0\}$$

$$\therefore \dim K(f) = 0$$

$$\therefore \dim [K(f)^\perp] = 0$$

$$\therefore \dim W = r$$

$$\therefore \dim [R(f|_W)] = r - 0 = r$$

$$\therefore R(f|_W) \subseteq W, \therefore f|_W \text{ 是满射}$$

$$\therefore \forall \beta \in W, \exists \eta \in W, \text{ 有 } \beta = f(\eta)$$

$$\therefore \forall \alpha \in W^\perp, \langle \alpha, \eta \rangle = 0 \text{ 满足}$$

$$f(\alpha), \beta = \langle f(\alpha), f(\eta) \rangle = \langle \alpha, \eta \rangle = 0$$

$$\therefore f(\alpha) \in W^\perp$$

$$\therefore W^\perp \text{ 是 } f \text{ 的不变子空间}$$

第3章

1. 设 $A \in C^{n \times n}$, $B \in C^{n \times s}$, 试证:

(1) $\text{tr} AB = \text{tr} BA$ (pp. 17)

(2) $\text{tr} (AB)^k = \text{tr} (BA)^k$, 其中 k 为正整数.

证: 令 $A = (a_{ij})_{n \times n}$, $B = (b_{ij})_{n \times s}$

$$\therefore \text{tr} AB = \sum_{i=1}^n \sum_{j=1}^s a_{ij} b_{ji}$$

$$\text{tr} BA = \sum_{j=1}^s \sum_{i=1}^n b_{ji} a_{ij}$$

$$\therefore \text{tr} AB = \text{tr} BA \quad (\text{以后可以直接用})$$

(3) $(AB)^k = \underbrace{ABAB \cdots AB}_k = AC$

$$(BA)^k = \underbrace{(BABA \cdots BA)}_k = CA$$

$$\therefore \text{tr} (AB)^k = \text{tr} (AC)$$

$$\text{tr} (BA)^k = \text{tr} (CA)$$

$$\text{由 (1)} \therefore \text{tr} AC = \text{tr} CA$$

$$\therefore \text{tr} (AB)^k = \text{tr} (BA)^k$$

2. 设 $A \in C^{n \times n}$, 存在正整数 k , 使 $A^k = 0$ 称 A 为幂零阵. 试证:

(1) $\det A = 0$;

(2) $\text{tr} A = 0$;

(3) $\det (I + A) = 1$.

4. 若 $A \neq 0$, 则 A 不能相似于对角阵.

(1) $|A| = |A|^k = 0 \therefore |A| = 0$

证: $\because A^k = 0 \therefore A$ 的特征值全为 0

$\therefore |A|$ 为 A 的特征值的乘积, 即为 0

(2) A 的迹等于其特征值之和.

$$\therefore \text{tr} A = 0$$

(3) $|\lambda I - A| = \lambda^n$

$$\therefore |\lambda I - (I + A)| = |(\lambda - 1)I - A| = (\lambda - 1)^n$$

$\therefore$ 矩阵 $I + A$ 的特征值全为 1.

$\therefore |I + A|$ 为 $(I + A)$ 特征值的乘积, 即为 1.

(4) 反证法.

若 A 相似于对角阵 $\Lambda = \begin{pmatrix} 0 & & \\ & 0 & \\ & & 0 \end{pmatrix} = 0$ (A 的特征值全为 0)

即 $\exists P$ 可逆, s.t. $A = P^{-1} \Lambda P = 0$ 与条件 $A \neq 0$ 矛盾.

$\therefore A$ 不能相似于对角阵.

4. 设 V 为 n 维内积空间, w 为 V 中单位向量. 作线性变换:

$$f(x) = x - 2(x, w)w \quad (x \in V) \quad (\text{pp. 161 镜像变换})$$

求 f 的特征多项式, 特征值及相应的特征子空间.

解: 将 w 扩充成 V 的一组标准正交基:

$$w, w_2, \dots, w_n. \quad f \text{ 在这组基下的矩阵为}$$

$$A = \begin{pmatrix} -1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$$

$$C(\lambda) = (\lambda + 1)(\lambda - 1)^{n-1}. \quad \text{特征值为 } -1, 1 \quad (n-1 \text{ 重})$$

$$V_{-1} = \{x \mid f(x) = -x, x \in V\}$$

$$\text{由 } f(w) = -w, w \text{ 为单位向量. } V_{-1} = L(w)$$

$$V_1 = \{x \mid f(x) = x, x \in V\}$$

$$\text{由 } f(w_i) = w_i \quad (i=2, \dots, n)$$

$$\therefore V_1 = L(w_2, \dots, w_n) = \{x \in V \mid x \perp w\} = L(w)^\perp$$

5. 设 $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, 求 $A^{100} - 3A^{25}$

解: $f(\lambda) = \lambda^{100} - 3\lambda^{25}$

$$C(\lambda) = C(\lambda)(\lambda - 1)^2$$

$$f(\lambda) = C(\lambda)q(\lambda) + a + b\lambda + c\lambda^2$$

$$f(A) = aI + bA + cA^2$$

$$f(1) = a + b + c = 4$$

$$f(1) = a + b + c = 2$$

$$f'(1) = b + 2c = 100 - 75 = 25$$

$$\Rightarrow \begin{cases} a = 13 \\ b = 3 \\ c = 14 \end{cases}$$

$$\therefore f(A) = 13I + 3A + 14A^2$$

$$= 13I + 3 \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} + 14 \begin{bmatrix} 5 & 0 & 2 \\ 4 & 1 & 2 \\ 8 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 54 & 6 & 25 \\ 50 & 2 & 25 \\ 112 & 12 & 52 \end{bmatrix}$$

不求解矩阵方程 $X^2 - X - 20I = 0$, 其中 $X \in C^{n \times n}$.

解: $\varphi(\lambda) = \lambda^2 - \lambda - 20 = (\lambda - 5)(\lambda + 4)$ 为 X 的特征多项式

$$\varphi(X) = 0 \quad \begin{cases} X \text{ 的特征值为 } 5 \text{ 或 } -4 \\ X \text{ 相似于对角阵} \end{cases}$$

(3) $\lambda \mid \varphi(\lambda)$, 而 $\varphi(\lambda)$ 为 λ 的二次多项式, λ 为重根.

$\therefore X$ 相似于对角阵.

$$X = P^{-1} \begin{pmatrix} 5I_r & \\ & -4I_{n-r} \end{pmatrix} P \quad P \text{ 可逆阵, } 1 \leq r \leq n.$$

13. 设 $A = (a_{ij})_{n \times n}$ 为上角阵且主对角元全等于 k
 试证: A 相似于对角阵的必要条件为 $A = kI$.

证: $\Leftarrow$ 若 $A = kI$ 则 A 为对角阵 $\checkmark$.

$$\Rightarrow: A \sim \Lambda = \begin{pmatrix} k & & \\ & k & \\ & & k \end{pmatrix} = kI.$$

$$\text{取可逆阵 } P, \text{ s.t. } A = P^{-1}(kI)P \\ = kP^{-1}IP \\ = kI.$$

$$\therefore A = kI.$$

15. 设 $J_\alpha = \begin{pmatrix} \alpha & 1 & & \\ & \alpha & \ddots & \\ & & \ddots & 1 \\ & & & \alpha \end{pmatrix} = \alpha I_k + N$, 且 $P(x)$ 为 A 的

多项式. 试证:

$$P(J_\alpha) = \begin{pmatrix} P(\alpha) & P'(\alpha) & \dots & P^{(k-1)}(\alpha)/(k-1)! \\ 0 & P(\alpha) & \dots & P^{(k-2)}(\alpha)/(k-2)! \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & P(\alpha) \end{pmatrix}_{k \times k}$$

证: $P(x)$ 在 A 展成 Taylor 级数.

$$P(x) = P(\alpha) + \frac{P'(\alpha)}{1!}(x-\alpha) + \frac{P''(\alpha)}{2!}(x-\alpha)^2 + \dots$$

$$P(J_\alpha) = P(\alpha)I + \frac{P'(\alpha)}{1!}(J_\alpha - \alpha I) + \frac{P''(\alpha)}{2!}(J_\alpha - \alpha I)^2 + \dots \\ = P(\alpha)I + \frac{P'(\alpha)}{1!}N + \frac{P''(\alpha)}{2!}N^2 + \dots \text{得证.}$$

16. 试写出满足下列条件的矩阵 A 的 Jordan 标准形之一切可能形式 (不计 Jordan 块的次序).

(1) A 的特征多项式为

$$C(\lambda) = (\lambda-a)^2(\lambda-b)^3 \quad (a \neq b);$$

(2) $C(\lambda) = (\lambda-a)^2(\lambda-b)^3$, 最小多项式.

$$m(\lambda) = (\lambda-a)(\lambda-b)^2 \quad (a \neq b);$$

(3) $C(\lambda) = (\lambda-a)^4$, $m(\lambda) = (\lambda-a)^2$, $r(A-aI) = 2$;

(4) $C(\lambda) = (\lambda-a)^7$, $m(\lambda) = (\lambda-a)^3$, $r(A-aI) = 4$;

解: (1) A 为 5 阶方阵.

$$J_A = \begin{pmatrix} a & * & & & \\ & a & & & \\ & & b & & \\ & & & b & * \\ & & & & b \end{pmatrix} \quad * = 0/1.$$

(2) 最小多项式反证若各块对应各特征值的最高阶数

$$J_\alpha = \begin{pmatrix} a & & 0 \\ & a & \\ & & b & \\ & & & b \end{pmatrix}$$

(3) $J_A = \begin{pmatrix} a & 1 & & \\ & a & & \\ & & a & \\ & & & a \end{pmatrix} \quad (r(A-aI)=2 \Rightarrow r(J_A-aI)=2)$

(4) $J_A = \begin{pmatrix} a & 1 & & & \\ & a & & & \\ & & a & & \\ & & & a & \\ & & & & a \end{pmatrix} \text{ 或 } \begin{pmatrix} a & 1 & & & \\ & a & & & \\ & & a & & \\ & & & a & \\ & & & & a \end{pmatrix}$

18. 试证: 复数域上任一 n 阶方阵 A 必有分解式

$$A = S + \Lambda$$

其中 S 可相似对角化, Λ 是幂零阵, 且 $S\Lambda = \Lambda S$.

证: 设 S 可逆阵 P , s.t. $P^{-1}AP = J = \begin{pmatrix} J_1 & & \\ & J_2 & \\ & & \ddots & \\ & & & J_s \end{pmatrix}$

$$J_i = \begin{pmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix}_{n_i \times n_i} = \lambda_i I_{n_i} + N_{n_i}.$$

$$\therefore J = \begin{pmatrix} \lambda_1 I_{n_1} & & \\ & \ddots & \\ & & \lambda_s I_{n_s} \end{pmatrix} + \begin{pmatrix} N_{n_1} & & \\ & \ddots & \\ & & N_{n_s} \end{pmatrix} \quad \sum_{i=1}^s n_i = n$$

$$= S + \Lambda, \text{ 显然 } S \text{ 为对角阵, } \Lambda \text{ 为幂零阵.}$$

$$\therefore A = P^{-1}JP = P^{-1}(S + \Lambda)P \\ = P^{-1}SP + P^{-1}\Lambda P \\ = S + \Lambda.$$

可验证这里 S 和 Λ 满足题设要求.

19. 已知 A 的 Jordan 标准形为 $J = \begin{pmatrix} a & 1 & & \\ & a & & \\ & & a & \\ & & & a \end{pmatrix}$.

试证: 当 $a \neq 0$, $a = 0$ 求 A^2 的 Jordan 标准形.

解: (1) $a \neq 0$: $A \sim J \Rightarrow A^2 \sim J^2$ 即求 J^2 的若当标准形.

$$J^2 = \begin{pmatrix} a^2 & 2a & & \\ & a^2 & 1 & \\ & & a^2 & \\ & & & a^2 \end{pmatrix} \quad \therefore r(J^2 - a^2 I) = 3.$$

$$\therefore A^2 \text{ 的若当标准形为 } \begin{pmatrix} a^2 & 1 & & \\ & a^2 & & \\ & & a^2 & \\ & & & a^2 \end{pmatrix}$$

(2) $a = 0$: $J^2 = \begin{pmatrix} 0 & 0 & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix} \quad r(J^2) = 2, \text{ 且 } (J^2)^2 = 0$

$$\therefore A^2 \text{ 的若当标准形为 } \begin{pmatrix} 0 & 1 & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix} \checkmark \quad \begin{pmatrix} 0 & 1 & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix} \times (J^2 = 0)$$

20. 设 α, β 为 n 维列向量, 研究矩阵 $\alpha\beta^H$ 的 Jordan 标准形.

解: $r(A) \leq (r(\alpha), r(\beta)) = 1$.

① $r(A) = 0 \Rightarrow (\alpha = 0 \text{ 或 } \beta = 0) \Rightarrow A = 0 \Rightarrow J_A = 0$.

$r(A) = 1$ A 的若当标准形 $r(J_A) = 1$.

$\therefore A$ 的若当标准形只有两种情形.

$J = \begin{pmatrix} * & 0 \\ 0 & 0 \end{pmatrix}$ 或 $K = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

$\text{tr}(A) \neq 0$ 时 $A \sim J$

$\text{tr}(A) = 0$ 时 $A \sim K$.

$\text{tr} A = \text{tr} \alpha\beta^H = \text{tr} \beta^H \alpha = \beta^H \alpha = \langle \beta, \alpha \rangle$

① 当 $\langle \alpha, \beta \rangle \neq 0$ 时 即 $\alpha \perp \beta$ 时 A 的若当标准形为 J 且 $P(A) \leq 10$.

② 当 $\langle \alpha, \beta \rangle = 0$ 即 $\alpha \perp \beta$ 时 A 的若当标准形为 K .

若 $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & a \end{pmatrix}$ 与 $B = \begin{pmatrix} a & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & b \end{pmatrix}$ 相似问:

a, b, c, d 应满足什么条件.

求 Jordan 标准形并求 P 使 $P^{-1}AP = J$.

(1) $\begin{bmatrix} 3 & 0 & 8 \\ 2 & -1 & 6 \\ -2 & 0 & -5 \end{bmatrix}$ (2) $\begin{bmatrix} 0 & 3 & 3 \\ 1 & 8 & 6 \\ 2 & -14 & -10 \end{bmatrix}$

解: (1) $C(\lambda) = (\lambda+1)^3 \sim J$

$r(A+I) = r(J+I) = 1$.

$\therefore A \sim J = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$.

(2) $C(\lambda) = \lambda(\lambda+1)^2$

A 可能的若当标准形有:

$J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ 和 $K = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

$r(A+I) = 2$.

$\therefore A \sim K$

求 P 使 $(\lambda - A)X = 0$ 各基础解系...

28. 设 $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{bmatrix}$, 试证: $P(A) = 10$.

证: 利用盖尔圆估计谱半径

$\therefore P(A) \leq p_1, p_2, p_1 = 10, p_2 = 10 \therefore P(A) \leq 10$

$r(A-10I) \leq 4, \therefore |A-10I| = 0$

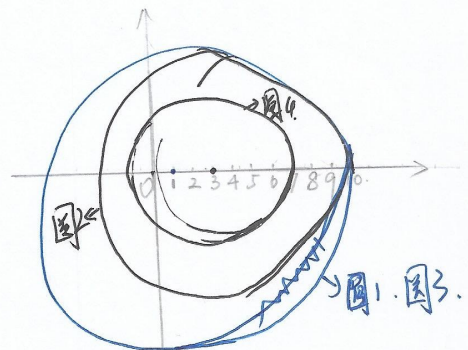
$\therefore 10$ 是 A 的一个特征值 $\therefore P(A) = 10$

或 $A \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = 10 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \therefore 10$ 为 A 的特征值 $\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ 为相应特征向量. $\therefore P(A) = 10$

29. 设 $A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{pmatrix}$; 试证: $P(A) < 10$

(证) 0 不是 A 的特征值不能说明 $P(A)$ 小于 10 而只是说明了 $(10, 0)$ 这一点不在以原点为圆心半径为 10 的圆上. 圆上还有另外的点.

找出所有盖尔圆:



离圆心最远的点为 $(10, 0)$ 这点. 也就是盖尔圆系中只有 $(10, 0)$ 这一点谱半径等于 10. 而 10 又不是特征值. 则 $P(A) < 10$.

30. 设 $A = \begin{bmatrix} 0 & \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} & \frac{1}{n} \\ \frac{1}{n} & 2 & \frac{1}{n} & \cdots & \frac{1}{n} & \frac{1}{n} \\ \frac{1}{n} & \frac{1}{n} & 4 & \cdots & \frac{1}{n} & \frac{1}{n} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \cdots & 2n-4 & \frac{1}{n} \\ \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} & 2n-2 \end{bmatrix}$ 试证: A 必相似于实对称阵.

解: 察思路: 要说明 A 的特征值互异且都是实数.
即 A 的盖尔圆均是 (互) 且圆心为实数.

A 的特征多项式为实系数多项式若有复数根.
例如 C 为 A 的特征多项式的根, 则其共轭复数 $\bar{C}$ 也为它的根, 与 A 的盖尔圆均是 (互) 的矛盾...

第4章

设 A 为正规阵. 试证:

(1) A 为 Hermite 阵 $\Leftrightarrow A$ 的特征值全为实数

(2) A 为酉阵 $\Leftrightarrow A$ 的特征值之模为 1.

证: $\because A$ 为正规阵.

$\therefore$ 存在矩阵 U , s.t. $U^H A U = \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$

$\lambda_1, \dots, \lambda_n$ 为 A 的所有特征值.

(1) " $\Rightarrow$ " $\checkmark$ Hermite 阵的每个元素全为实数. 特征值 λ_i

$\therefore A = U \Lambda U^H$

$A^H = U \Lambda^H U^H$

$\because \lambda_i = \bar{\lambda}_i$ 全为实数.

$\therefore \Lambda^H = \Lambda$

$\therefore A^H = U \Lambda U^H = A$

$\therefore A$ 为 Hermite 阵.

(2) " $\Rightarrow$ " $\because A$ 为酉阵.

$\therefore \Lambda = U^H A U$ 为酉阵.

$\therefore \Lambda^H = I$

$\therefore \forall \lambda_i, \bar{\lambda}_i \lambda_i = \|\lambda_i\|^2 = 1. \therefore \|\lambda_i\| = 1.$

6. 设 $\alpha \in \mathbb{C}^n$, 且 $\|\alpha\| < 1$. 试证: $I - \alpha \alpha^H$ 是正定阵.

证: 令 $A = I - \alpha \alpha^H$.

$\therefore A^H = A$

$\therefore A$ 为 Hermite 阵.

$\because \alpha \alpha^H$ 的特征值为: $\alpha^H \alpha$ 和 $0 (n-1 \text{ 重})$

$\therefore A$ 的特征值为: $1 - \alpha^H \alpha, 1 (n-1 \text{ 重})$

$\because \alpha^H \alpha < 1$.

$\therefore A$ 的特征值全大于 0.

$\therefore A = I - \alpha \alpha^H$ 是正定阵.

8. 试证 Hermite 阵 A 是半正定的充要条件是 A 的一切主子式非负.

证: $A = S^H S$.

$|A| = |S^H| \cdot |S| = \|S\|^2 \geq 0 \dots$

$\Leftarrow$ 要证明 A 的特征值非负.

$$C_A(\lambda) = \lambda^n + (-1)^1 b_1 \lambda^{n-1} + (-1)^2 b_2 \lambda^{n-2} + \dots + (-1)^n b_n \lambda^0 + (-1)^n b_n.$$

$b_i = \sum (\text{A 的 } i \text{ 阶主子式}) \geq 0$

$$C_A(\lambda I - A) = \lambda^n - \underbrace{b_1}_{<0} \lambda^{n-1} + \underbrace{b_2}_{>0} \lambda^{n-2} - \underbrace{b_3}_{<0} \lambda^{n-3} + \dots$$

这样的多项式的根都大于 0 (根为实数: 由韦达)

$\therefore A$ 半正定.

9. 试证: Hermite 阵 A 是负定的 $\Leftrightarrow A$ 的所有主子式与 $(-1)^k$ 同号.

证: $A = (a_{ij})_{n \times n}$ 负定 $\Leftrightarrow -A$ 正定.

$-A = \begin{pmatrix} -a_{11} & -a_{12} & \dots & -a_{1n} \\ -a_{21} & -a_{22} & \dots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \dots & -a_{nn} \end{pmatrix}$ 正定.

$\Leftrightarrow -a_{11} > 0 \quad a_{11} < 0.$

$\begin{vmatrix} -a_{11} & -a_{12} \\ -a_{21} & -a_{22} \end{vmatrix} > 0 \quad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0$

$-A$ 的 k 阶主子式 > 0 A 的 k 阶主子式 $= (-1)^k \cdot (-A)$ 的 k 阶主子式 > 0 与 $(-1)^k$ 同号.

11. 设 A 为半正定. 试证:

(1) 必存在半正定阵 S , 使 $A = S^2$.

(2) $|Y^H A X|^2 \leq |Y^H A Y| |X^H A X|, \forall X, Y \in \mathbb{C}^n$

(1) 存在酉阵 U , s.t. $U^H A U = \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \quad \lambda_i \geq 0$

$A = U \Lambda U^H$ 令 $\Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$

$= U \Lambda U^H = U \Lambda U^H U \Lambda U^H = (U \Lambda U^H)^2$

令 $S = U \Lambda U^H$ 为 Hermite 阵.

S 半正定. S 由 λ_i 决定

$\therefore A = S^2$.

12). 由 (1) $A = S^2$ (S 半正定)
 $Y^H A X = Y^H S \cdot S X = Y^H S^H \cdot S X$
 $= (SY)^H (SX)$
 $= \langle SX, SY \rangle$

由 C-B 不等式:

$$\begin{aligned} |Y^H A X|^2 &= |\langle SX, SY \rangle|^2 \\ &\leq |SX|^2 |SY|^2 \\ &= (SX)^H SX (SY)^H SY \\ &= X^H S^H S X \cdot Y^H S^H S Y \\ &= X^H A X \cdot Y^H A Y \end{aligned}$$

B. 设 A, B 为同阶 Hermit 矩阵, 且 A 为正定阵.

试证存在可逆矩阵 C , s.t. $C^H A C$ 与 $C^H B C$ 均为对角阵.

证: $\because A$ 正定.

$\therefore \exists$ 可逆阵 P , s.t. $P^H A P = I$

$$\& (P^H B P)^H = P^H B^H P = P^H B P$$

$\therefore P^H B P$ 为 Hermit 阵.

$\therefore \exists$ 酉交阵 U , s.t. $U^H (P^H B P) U = \Lambda$ 对角阵.

$$U^H (P^H A P) U = I.$$

令 $C = PU$ 可逆. 得证.

4. 设 A 为 n 阶阵, 作 n 元 Hermit 阵二次型

$$f(x) = \begin{vmatrix} A & x \\ x^H & 0 \end{vmatrix}, \forall x \in C^n.$$

试证: f 是实数.

$$\begin{pmatrix} I & 0 \\ -x^H A^H & 1 \end{pmatrix} \begin{pmatrix} A & x \\ x^H & 0 \end{pmatrix} = \begin{pmatrix} A & x \\ 0 & -x^H A^H x \end{pmatrix}$$

$$\text{取行列式: 左边} = f(x) = -|A| \det A^H x$$

$\because A$ 正定.

$\therefore -|A| < 0$. A 正定, $x^H A^H x$ 为实数.

$\therefore f(x)$ 为实数.

11. A 为同阶 Hermit 阵, 且 A 为正定阵. 试证:
 AB 的特征值均为实数.

$\because A$ 正定.

$\therefore \exists$ 可逆阵 S s.t. $A = S^H P S$.

$$AB = S^H P B S.$$

$$\therefore (P^H)^{-1} A B P^H \sim AB.$$

$$= (P^H)^{-1} P^H P B P^H = P B P^H \sim AB$$

$$\therefore (P B P^H)^H = P B^H P^H = P B P^H$$

$\therefore P B P^H$ 为 Hermit 阵.

$\therefore AB$ 为 Hermit 阵.

$\therefore AB$ 的特征值均为实数.

$$P^H A P = I.$$

$$A = (P^H)^{-1} I P^H$$

$$= (P^H)^{-1} I P^H$$

$$= P^H I P^H$$

2. 试证: n 阶阵 A 为正规阵的充分必要条件是

$$\|AX\| = \|A^H X\| \quad (\forall X \in C^n)$$

$$\text{证: } \|AX\| = \|A^H X\|$$

$$\Leftrightarrow \langle AX, AX \rangle = \langle A^H X, A^H X \rangle$$

$$\Leftrightarrow (AX)^H AX = (A^H X)^H A^H X$$

$$\Leftrightarrow X^H A^H A X = X^H A A^H X$$

$$\Leftrightarrow X^H (A^H A - A A^H) X = 0 \quad \forall X \in C^n.$$

$$\Leftrightarrow A^H A - A A^H = 0$$

$$\Leftrightarrow A^H A = A A^H \quad \text{即 } A \text{ 为正规阵}$$

以上步骤可逆, 故为充要条件

3. 试证: (1) 若 A 为正规阵, 则 $A - \lambda I$ 也是正规阵;

(2) 若 A 为正规阵, 则 $AX = \lambda X$ 之充分必要条件为

$$A^H X = \bar{\lambda} X$$

(即 A 的特征值 λ 之共轭 $\bar{\lambda}$ 为 A^H 的特征值且对应于相同特征向量).

$$\text{证: (1) } (A - \lambda I)^H (A - \lambda I) = (A^H - \bar{\lambda} I) (A - \lambda I)$$

$$= A^H A - \lambda A^H - \bar{\lambda} A + \lambda \bar{\lambda} I$$

$$= A^H A - \lambda A^H - \bar{\lambda} A + |\lambda|^2 I \quad \text{①}$$

$$(A - \lambda I) (A - \lambda I)^H = (A - \lambda I) (A^H - \bar{\lambda} I)$$

$$= A A^H - \bar{\lambda} A - \lambda A^H + |\lambda|^2 I \quad \text{②}$$

$$\because A \text{ 为正规阵 } \therefore A A^H = A^H A$$

$$\therefore \text{①式} = \text{②式} \therefore A - \lambda I \text{ 为正规阵.}$$

证: (1) $AX = \lambda X$. 即 A 的特征值为 λ , 相应
特征向量为 X .

$$(A - \lambda I)X = 0.$$

$\because A$ 为正规阵.

由 (1) $A - \lambda I$ 也为正规阵.

$$\text{故由 } 2. \| (A - \lambda I)X \| = \| (A - \lambda I)^H X \| = 0$$

$$\therefore (A - \lambda I)^H X = 0$$

$$\therefore A^H X = \bar{\lambda} X.$$